

1. Given the dataset in the table below, use the Naive Bayes classifier to classify the new point (F, T, 2.0).

Assume that no Laplace smoothing is applied. Also assume that a_3 is distributed for class Y and class N using normal distributions with different μ and σ , and that these parameters are estimated from the given data using maximum likelihood estimation.

x_i	a_1	a_2	a_3	class
x_1	T	F	6.0	Y
x_2	F	T	7.5	Y
x_3	T	T	5.5	N
x_4	F	F	3.5	Y
x_5	T	F	8.5	N
x_6	F	T	4.5	N
x_7	T	T	6.5	Y
x_8	F	F	5.0	N
x_9	T	F	7.0	Y

2. Consider the following training points:

$$(2, 1, +), (1, -1, -), (0, 2, +), (-1, 0, -)$$

Check whether the data are linearly separable or not. If not, then transform the data from input space to feature space using the following transformation function ϕ .

Find the maximum margin classifier (linear decision boundary) in the transformed space (assume hard-margin SVM).

$$\phi = (1, \sqrt{3}x_1, \sqrt{3}x_2, \sqrt{3}x_1x_2, x_1^2, x_2^2)$$

3. Consider the following dataset:

city	job	credit limit	age	class
Delhi	employed	2-3 lakhs	25	yes
Mumbai	student	1-2 lakhs	19	no
Delhi	unemployed	1 lakh	40	no
Kolkata	employed	3-4 lakhs	35	yes
Mumbai	employed	2-3 lakhs	45	yes
Kolkata	student	1 lakh	22	no
Delhi	employed	3-4 lakhs	30	yes
Mumbai	unemployed	1-2 lakhs	28	no

- (a) Compute the entropy of the dataset and the information gain for all attributes.

- (b) Compute the Gini index for all attributes and identify the best attribute for the root node.
4. Consider a feedforward neural network with one hidden layer.
The input vector is:

$$p = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T = 1 \text{ (target output)}$$

The network uses Sigmoid activation function in the hidden layer, given by $\sigma(x) = \frac{1}{1+e^{-x}}$, and Linear activation function in the output layer. Let y denote the final output of the network.

The initial weights and biases are:

$$W^1 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad b^1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$W^2 = [1 \quad 2], \quad b^2 = 0$$

- (a) Perform the forward propagation and compute the output y .
(b) Compute the error using:

$$E = \frac{1}{2}(T - y)^2$$

- (c) Perform one iteration of backpropagation with learning rate $\alpha = 0.1$.
(d) Update all weights and biases.
5. Consider a logistic regression model for binary classification. The hypothesis function is given by:
 $h_{\theta}(x) = \frac{1}{1+e^{-z}}$, where $z = \theta_0 + \theta_1 x$
Given: $\theta_0 = -1, \theta_1 = 0.5$ and a data point: $x = 4, y = 1$
- (a) Compute the value of z .
(b) Compute the predicted probability $h_{\theta}(x)$.
(c) Compute the cost using the logistic loss function: $J = -[y \log(h_{\theta}(x)) + (1 - y) \log(1 - h_{\theta}(x))]$
(d) Compute the gradient of the cost with respect to θ_0 and θ_1 .
(e) Perform one step of gradient descent with learning rate $\alpha = 0.1$ and update the parameters.
6. Consider the dataset shown in Table 1. Estimate the conditional probabilities $P(A|+)$, $P(B|+)$, $P(C|+)$, $P(A|-)$, $P(B|-)$, and $P(C|-)$.
For the instance $X = (A = 1, B = 0, C = 1)$, compute the posterior probabilities $P(+|X)$ and $P(-|X)$.

Record	A	B	C	Class
1	1	0	1	+
2	0	1	0	-
3	1	1	1	+
4	0	0	0	-
5	1	0	0	+
6	0	1	1	-
7	1	1	0	+
8	0	0	1	-
9	1	0	1	+
10	0	1	0	-

7. Given the points

$P_1(4, 8)$, $P_2(5, 6)$, $P_3(6, 5)$, $P_4(7, 5)$, $P_5(6, 4)$, $P_6(8, 6)$, $P_7(10, 3)$, $P_8(11, 3)$

use the DBSCAN algorithm (with $\epsilon = 2$, MinPts = 3, and Manhattan distance) to classify every point as a core point, border point, or outlier.

Also, state the final resulting clusters.

8. Perform one iteration of K-Means clustering algorithm. Choose $K = 2$, use Euclidean distance. Choose first and last data points as initial centers.

Sl. No	Point	Coordinates
1	A	(1, 1)
2	B	(2, 1)
3	C	(4, 3)
4	D	(5, 4)
5	E	(3, 2)

9. Given the following training dataset:

Point	Coordinates (x_1, x_2)	Class
A	(1, 2)	+
B	(2, 3)	+
C	(5, 6)	-
D	(6, 7)	-

A query point $q = (3, 3)$ is given.

- Compute the Euclidean distance between the query point and each of the training points.
- Using the K-Nearest Neighbors algorithm, determine the predicted class label of the query point for:

- (i) $k = 1$
 - (ii) $k = 3$
 - (c) Explain briefly why the predicted class may change when the value of k is increased.
10. An agent runs 3 episodes in a maze. The returns observed for state s_2 in each episode (assume returns are already computed using discounting) are:
- Episode 1: $G = 8$ (visited twice, at $t = 2$ and $t = 5$), Episode 2: $G = 4$ (visited once, at $t = 1$), Episode 3: $G = 10$ (visited once, at $t = 3$).
- (a) Compute $V(s_2)$ using first-visit Monte Carlo (MC) and every-visit MC separately.
 - (b) Under what conditions do both methods converge to the same value? Provide a formal argument.
 - (c) Why is Monte Carlo defined only for episodic tasks, and what is the consequence of applying it to a continuing task?
11. Given the following 2D dataset:

Point	Coordinates (x, y)	Class label $(y = \pm 1)$
A	(1, 1)	+1
B	(2, 2)	+1
C	(-1, -1)	-1
D	(-2, -2)	-1

- (a) Sketch or describe the positions of these points in the 2D plane. Are the data points linearly separable? Justify your answer and identify a possible separating line.
 - (b) Determine the equation of the maximum-margin separating hyperplane in the form:
$$\mathbf{w} \cdot \mathbf{x} + b = 0$$

Compute the values of $\mathbf{w}$ and b , and calculate the margin width. Show all necessary steps.
 - (c) Identify the support vectors among the points A, B, C, and D. Justify your answer.
12. (a) Explain the concept of the kernel trick in Support Vector Machines (SVM). Why does the use of a kernel function eliminate the need to explicitly compute a mapping to a higher-dimensional feature space?
- (b) Consider the polynomial kernel of degree 2 defined as:

$$K(\mathbf{x}, \mathbf{z}) = (1 + \mathbf{x}^\top \mathbf{z})^2$$

Find an explicit feature mapping $\phi(\mathbf{x})$ such that:

$$K(\mathbf{x}, \mathbf{z}) = \phi(\mathbf{x})^\top \phi(\mathbf{z})$$

For a 2-dimensional input $\mathbf{x} = (x_1, x_2)$, write all components of the transformed feature vector $\phi(x_1, x_2)$.

(c) For the points:

$$\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{z} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

- (i) Compute the kernel value $K(\mathbf{x}, \mathbf{z})$.
- (ii) Using the feature mapping obtained in part (b), compute $\phi(\mathbf{x})^\top \phi(\mathbf{z})$.
- (iii) Verify that the two results are equal.